

AERIS Expert Review Packet

Anomaly

Source / metric	tceq / Nitrogen dioxide (no2)
Observed / expected baseline	44.3 / 6.78264 ppb
Baseline method	mean of the preceding 7 days of this series (Z-score detector)
Time	2026-07-22 02:00 UTC
Location	29.8027, -95.1255
Severity	severe
Detectors	zscore, stl, isolation_forest

Stored evidence summary

The three tables in this section are look-up tables. You never have to read them through: they are here for the moment a claim quotes a number and you want to check it.

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
asos	Relative humidity (humidity)	percent	7 / 724	29.01 to 100; mean 65.7339	55.64 at 2026-07-22 01:55Z; 5 min before event
asos	Air temperature (temperature)	degC	7 / 724	23.2222 to 39; mean 30.9161	32 at 2026-07-22 01:55Z; 5 min before event
asos	Wind direction (wind_direction)	degree	7 / 886	0 to 360; mean 193.33	210 at 2026-07-22 01:55Z; 5 min before event
asos	Wind speed (wind_speed)	m/s	7 / 923	0 to 8.2311; mean 2.8247	2.05778 at 2026-07-22 01:55Z; 5 min before event
noaa_gfs	500-hPa geopotential height (gh_500)	m	9 / 108	5915.17 to 5948.41; mean 5929.78	5929.88 at 2026-07-22 00:00Z; 2 h before event
noaa_gfs	Planetary boundary layer height (pbl_height)	m	9 / 108	17.7446 to 2110.42; mean 674.191	1535.76 at 2026-07-22 00:00Z; 2 h before event
noaa_gfs	Precipitable water (precipitable_water)	mm	9 / 108	29.7729 to 67.0225; mean 43.4465	40.8356 at 2026-07-22 00:00Z; 2 h before event
noaa_gfs	Surface pressure (surface_pressure)	hPa	9 / 108	1002.85 to 1013.76; mean 1008.65	1007.88 at 2026-07-22 00:00Z; 2 h before event
noaa_gfs	850-hPa temperature (t_850)	degC	9 / 108	19.888 to 24.7709; mean 22.0802	21.8917 at 2026-07-22 00:00Z; 2 h before event
noaa_gfs	10-m eastward wind (u_10m)	m/s	9 / 108	-4.5306 to 4.0994; mean 1.1172	1.91716 at 2026-07-22 00:00Z; 2 h before event
noaa_gfs	10-m northward wind (v_10m)	m/s	9 / 108	-4.10555 to 4.8277; mean -0.2757	-1.06296 at 2026-07-22 00:00Z; 2 h before event
openaq	Ozone (ozone)	ppm	16 / 1030	-0.004 to 0.094; mean 0.0282	0.042 at 2026-07-22 02:00Z; at event time

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
openaq	PM10 (pm10)	ug/m3	3 / 160	23 to 98; mean 44.3813	53 at 2026-07-22 02:00Z; at event time
openaq	PM2.5 (pm25)	ug/m3	76 / 2796	0.3 to 31.2; mean 8.8891	17.5 at 2026-07-22 02:00Z; at event time
openweather	Cloud cover (cloud_cover)	percent	5 / 359	0 to 100; mean 22.9109	4 at 2026-07-22 01:59Z; at event time
openweather	Relative humidity (humidity)	percent	5 / 359	40 to 95; mean 68.6657	52 at 2026-07-22 01:59Z; at event time
openweather	Precipitation rate (precipitation)	mm/h	5 / 359	0 to 11.53; mean 0.1179	0 at 2026-07-22 01:59Z; at event time
openweather	Surface pressure (pressure)	hPa	5 / 359	1007 to 1015; mean 1011.02	1010 at 2026-07-22 01:59Z; at event time
openweather	Air temperature (temperature)	degC	5 / 359	24.14 to 40.07; mean 31.6508	33.62 at 2026-07-22 01:59Z; at event time
openweather	Wind direction (wind_direction)	degree	5 / 359	0 to 360; mean 204.822	290 at 2026-07-22 01:59Z; at event time
openweather	Wind speed (wind_speed)	m/s	5 / 359	0.4 to 6.11; mean 2.1085	1.79 at 2026-07-22 01:59Z; at event time
purpleair	PM2.5 (pm25)	ug/m3	66 / 4725	0 to 149.3; mean 10.7511	14 at 2026-07-22 01:59Z; at event time
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	count	6 / 6	1 to 1; mean 1	1 at 2026-07-21 19:34Z; 6 h 25 min before event
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	count	6 / 6	1 to 1; mean 1	1 at 2026-07-21 19:34Z; 6 h 25 min before event
sentinel5p	CO column (s5p_co_column)	mol/m^2	6 / 6	0.0262761 to 0.0312825; mean 0.028	0.0265506 at 2026-07-21 19:34Z; 6 h 25 min before event
sentinel5p	CO granules available (s5p_co_granule_available)	count	6 / 6	1 to 1; mean 1	1 at 2026-07-21 19:34Z; 6 h 25 min before event
sentinel5p	HCHO column (s5p_hcho_column)	mol/m^2	5 / 5	0.000212265 to 0.000396712; mean 0.0003	0.000224886 at 2026-07-21 19:34Z; 6 h 25 min before event
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	count	5 / 5	1 to 1; mean 1	1 at 2026-07-21 19:34Z; 6 h 25 min before event
sentinel5p	NO2 column (s5p_no2_column)	mol/m^2	3 / 3	5.0358e-05 to 5.87309e-05; mean 0.0001	5.87309e-05 at 2026-07-21 19:34Z; 6 h 25 min before event
sentinel5p	NO2 granules available (s5p_no2_granule_available)	count	3 / 3	1 to 1; mean 1	1 at 2026-07-21 19:34Z; 6 h 25 min before event
sentinel5p	O3 granules available (s5p_o3_granule_available)	count	6 / 6	1 to 1; mean 1	1 at 2026-07-21 19:34Z; 6 h 25 min before event
sentinel5p	SO2 column (s5p_so2_column)	mol/m^2	5 / 5	-8.60324e-06 to 4.32022e-05; mean 0	3.4166e-05 at 2026-07-21 19:34Z; 6 h 25 min before event
sentinel5p	SO2 granules available (s5p_so2_granule_available)	count	5 / 5	1 to 1; mean 1	1 at 2026-07-21 19:34Z; 6 h 25 min before event
tceq	Carbon monoxide (co)	ppm	3 / 192	0 to 0.5; mean 0.1474	0.2 at 2026-07-22 02:00Z; at event time
tceq	Nitrogen dioxide (no2)	ppb	12 / 701	-2.6 to 44.3; mean 6.2304	44.3 at 2026-07-22 02:00Z; at event time

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
tceq	Sulfur dioxide (so2)	ppb	3 / 186	-0.6 to 6.9; mean 0.0753	0.4 at 2026-07-22 02:00Z; at event time

Six-hour averages

Each metric averaged in 6-hour blocks across the stored 72 hours, in UTC. These averages, and the per-site ones below, are what the models were shown.

Source	Metric	Values
asos	Relative humidity (humidity)	07-20 12Z 68.22, 18Z 48.76, 07-21 00Z 75.65, 06Z 92.16, 12Z 73.32, 18Z 39.33, 07-22 00Z 61.22, 06Z 78.48, 12Z 59.36, 18Z 42.31, 07-23 00Z 69.74, 06Z 75.93, 12Z 75.56
asos	Air temperature (temperature)	07-20 12Z 30.32, 18Z 34.4, 07-21 00Z 29.33, 06Z 25.29, 12Z 29.09, 18Z 35.47, 07-22 00Z 30.65, 06Z 27.1, 12Z 31.74, 18Z 37.83, 07-23 00Z 31.46, 06Z 29.11, 12Z 28.61
asos	Wind direction (wind_direction)	07-20 12Z 245.5, 18Z 159.2, 07-21 00Z 178.9, 06Z 199.3, 12Z 261.2, 18Z 229.4, 07-22 00Z 186.8, 06Z 239.9, 12Z 289.3, 18Z 213.3, 07-23 00Z 141.1, 06Z 58.29, 12Z 23.08
asos	Wind speed (wind_speed)	07-20 12Z 2.839, 18Z 2.209, 07-21 00Z 2.984, 06Z 2.261, 12Z 3.595, 18Z 2.414, 07-22 00Z 2.254, 06Z 2.931, 12Z 3.799, 18Z 2.691, 07-23 00Z 2.144, 06Z 2.986, 12Z 5.105
noaa_gfs	500-hPa geopotential height (gh_500)	07-20 18Z 5928, 07-21 00Z 5923, 06Z 5926, 12Z 5917, 18Z 5932, 07-22 00Z 5930, 06Z 5935, 12Z 5927, 18Z 5939, 07-23 00Z 5930, 06Z 5943, 12Z 5927
noaa_gfs	Planetary boundary layer height (pbl_height)	07-20 18Z 1279, 07-21 00Z 797.8, 06Z 214.9, 12Z 211.9, 18Z 1233, 07-22 00Z 848.5, 06Z 119.6, 12Z 230.7, 18Z 1782, 07-23 00Z 568.3, 06Z 418.5, 12Z 385.9
noaa_gfs	Precipitable water (precipitable_water)	07-20 18Z 32.95, 07-21 00Z 36.38, 06Z 34.05, 12Z 32.13, 18Z 36.12, 07-22 00Z 40.71, 06Z 45.51, 12Z 52.33, 18Z 55.52, 07-23 00Z 60.01, 06Z 45.68, 12Z 49.97
noaa_gfs	Surface pressure (surface_pressure)	07-20 18Z 1012, 07-21 00Z 1009, 06Z 1009, 12Z 1011, 18Z 1010, 07-22 00Z 1008, 06Z 1008, 12Z 1009, 18Z 1008, 07-23 00Z 1005, 06Z 1007, 12Z 1008
noaa_gfs	850-hPa temperature (t_850)	07-20 18Z 22.3, 07-21 00Z 21.31, 06Z 23.01, 12Z 22.36, 18Z 22.24, 07-22 00Z 21.8, 06Z 22.47, 12Z 21.74, 18Z 21.82, 07-23 00Z 24.04, 06Z 21.66, 12Z 20.21
noaa_gfs	10-m eastward wind (u_10m)	07-20 18Z 1.846, 07-21 00Z -0.4616, 06Z 1.84, 12Z 2.68, 18Z 2.347, 07-22 00Z 1.118, 06Z 2.325, 12Z 3.049, 18Z 0.5087, 07-23 00Z 0.07829, 06Z -1.399, 12Z -0.5244
noaa_gfs	10-m northward wind (v_10m)	07-20 18Z 0.614, 07-21 00Z 2.21, 06Z 2.33, 12Z 0.296, 18Z -0.6915, 07-22 00Z -0.03407, 06Z 1.206, 12Z -0.5225, 18Z -2.34, 07-23 00Z 0.1966, 06Z -3.088, 12Z -3.486
openaq	Ozone (ozone)	07-20 12Z 0.01462, 18Z 0.04376, 07-21 00Z 0.01854, 06Z 0.006247, 12Z 0.01129, 18Z 0.04487, 07-22 00Z 0.02856, 06Z 0.006269, 12Z 0.0158, 18Z 0.06603, 07-23 00Z 0.0389, 06Z 0.02701, 12Z 0.02498
openaq	PM10 (pm10)	07-20 12Z 49.25, 18Z 50.75, 07-21 00Z 43.55, 06Z 39.83, 12Z 41.33, 18Z 49.09, 07-22 00Z 46.83, 06Z 42.2, 12Z 45.33, 18Z 51.67, 07-23 00Z 55, 06Z 30.56, 12Z 29.11
openaq	PM2.5 (pm25)	07-20 12Z 6.582, 18Z 6.645, 07-21 00Z 6.822, 06Z 7.454, 12Z 7.045, 18Z 5.732, 07-22 00Z 6.19, 06Z 6.086, 12Z 6.312, 18Z 9.198, 07-23 00Z 10.86, 06Z 18.66, 12Z 18.4
openweather	Cloud cover (cloud_cover)	07-20 12Z 2.5, 18Z 2.967, 07-21 00Z 3.69, 06Z 0.3226, 12Z 1.138, 18Z 1.806, 07-22 00Z 3.267, 06Z 21.53, 12Z 11.6, 18Z 56.93, 07-23 00Z 76.31, 06Z 66.06, 12Z 91
openweather	Relative humidity (humidity)	07-20 12Z 74.65, 18Z 53.33, 07-21 00Z 71.24, 06Z 89.23, 12Z 79.59, 18Z 48, 07-22 00Z 61.2, 06Z 79.5, 12Z 70.5, 18Z 49.03, 07-23 00Z 70.03, 06Z 76.65, 12Z 79.56
openweather	Precipitation rate (precipitation)	07-20 12Z 0, 18Z 0, 07-21 00Z 0, 06Z 0, 12Z 0, 18Z 0, 07-22 00Z 0, 06Z 0, 12Z 0, 18Z 0.006, 07-23 00Z 1.453, 06Z 0, 12Z 0
openweather	Surface pressure (pressure)	07-20 12Z 1015, 18Z 1012, 07-21 00Z 1012, 06Z 1012, 12Z 1013, 18Z 1011, 07-22 00Z 1010, 06Z 1010, 12Z 1011, 18Z 1008, 07-23 00Z 1009, 06Z 1010, 12Z 1010
openweather	Air temperature (temperature)	07-20 12Z 30.39, 18Z 35.79, 07-21 00Z 30.97, 06Z 25.89, 12Z 28.79, 18Z 36.46, 07-22 00Z 31.98, 06Z 27.45, 12Z 31.06, 18Z 38.97, 07-23 00Z 32.77, 06Z 29.84, 12Z 28.48
openweather	Wind direction (wind_direction)	07-20 12Z 238.7, 18Z 215.6, 07-21 00Z 183.5, 06Z 198.9, 12Z 249.6, 18Z 259.8, 07-22 00Z 234.3, 06Z 189.3, 12Z 255.2, 18Z 179.6, 07-23 00Z 208.2, 06Z 107.8, 12Z 42
openweather	Wind speed (wind_speed)	07-20 12Z 1.921, 18Z 1.766, 07-21 00Z 2.368, 06Z 2.22, 12Z 2.04, 18Z 1.646, 07-22 00Z 2.314, 06Z 1.842, 12Z 1.942, 18Z 2.202, 07-23 00Z 2.64, 06Z 2.198, 12Z 2.687

Source	Metric	Values
purpleair	PM2.5 (pm25)	07-20 12Z 8.996, 18Z 7.676, 07-21 00Z 8.884, 06Z 9.611, 12Z 8.515, 18Z 7.786, 07-22 00Z 9.78, 06Z 6.863, 12Z 7.975, 18Z 11.13, 07-23 00Z 14.64, 06Z 22.71, 12Z 23.23
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	07-20 18Z 1, 07-21 18Z 1, 07-22 18Z 1
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	07-20 18Z 1, 07-21 18Z 1, 07-22 18Z 1
sentinel5p	CO column (s5p_co_column)	07-20 18Z 0.02628, 07-21 18Z 0.02656, 07-22 18Z 0.03123
sentinel5p	CO granules available (s5p_co_granule_available)	07-20 18Z 1, 07-21 18Z 1, 07-22 18Z 1
sentinel5p	HCHO column (s5p_hcho_column)	07-20 18Z 0.000213, 07-21 18Z 0.0002243, 07-22 18Z 0.0003967
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	07-20 18Z 1, 07-21 18Z 1, 07-22 18Z 1
sentinel5p	NO2 column (s5p_no2_column)	07-20 18Z 5.835e-05, 07-21 18Z 5.873e-05, 07-22 18Z 5.036e-05
sentinel5p	NO2 granules available (s5p_no2_granule_available)	07-20 18Z 1, 07-21 18Z 1, 07-22 18Z 1
sentinel5p	O3 granules available (s5p_o3_granule_available)	07-20 18Z 1, 07-21 18Z 1, 07-22 18Z 1
sentinel5p	SO2 column (s5p_so2_column)	07-20 18Z 1.73e-05, 07-21 18Z 3.63e-05, 07-22 18Z -2.412e-06
sentinel5p	SO2 granules available (s5p_so2_granule_available)	07-20 18Z 1, 07-21 18Z 1, 07-22 18Z 1
tceq	Carbon monoxide (co)	07-20 12Z 0.16, 18Z 0.1722, 07-21 00Z 0.2222, 06Z 0.1222, 12Z 0.1556, 18Z 0.1056, 07-22 00Z 0.21, 06Z 0.1056, 12Z 0.1556, 18Z 0.1222, 07-23 00Z 0.18, 06Z 0.1333, 12Z 0.1667
tceq	Nitrogen dioxide (no2)	07-20 12Z 5.729, 18Z 6.136, 07-21 00Z 6.723, 06Z 5.171, 12Z 5.402, 18Z 4.761, 07-22 00Z 10.57, 06Z 6.367, 12Z 7.011, 18Z 6.311, 07-23 00Z 7.105, 06Z 5.303, 12Z 6.991
tceq	Sulfur dioxide (so2)	07-20 12Z -0.12, 18Z 1.906, 07-21 00Z 1.213, 06Z -0.2778, 12Z -0.275, 18Z -0.1611, 07-22 00Z -0.1222, 06Z -0.2889, 12Z -0.07222, 18Z 0.07778, 07-23 00Z -0.1667, 06Z -0.3, 12Z -0.1667

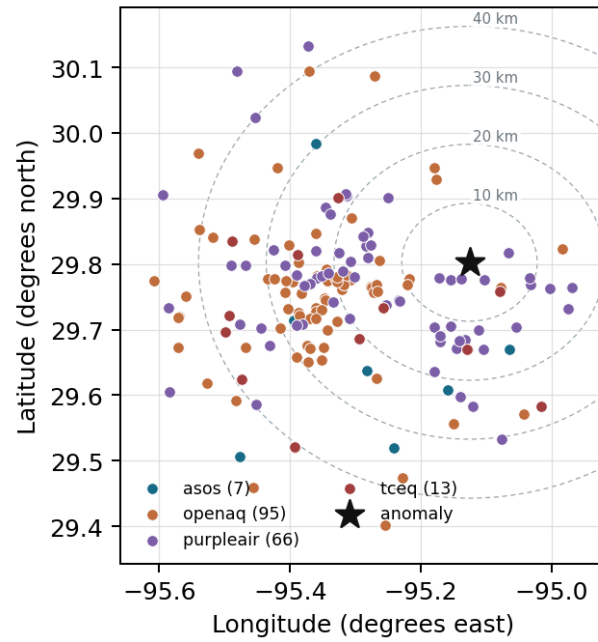
Per-site averages

Each metric averaged at each site, with that site's distance from the event in brackets.

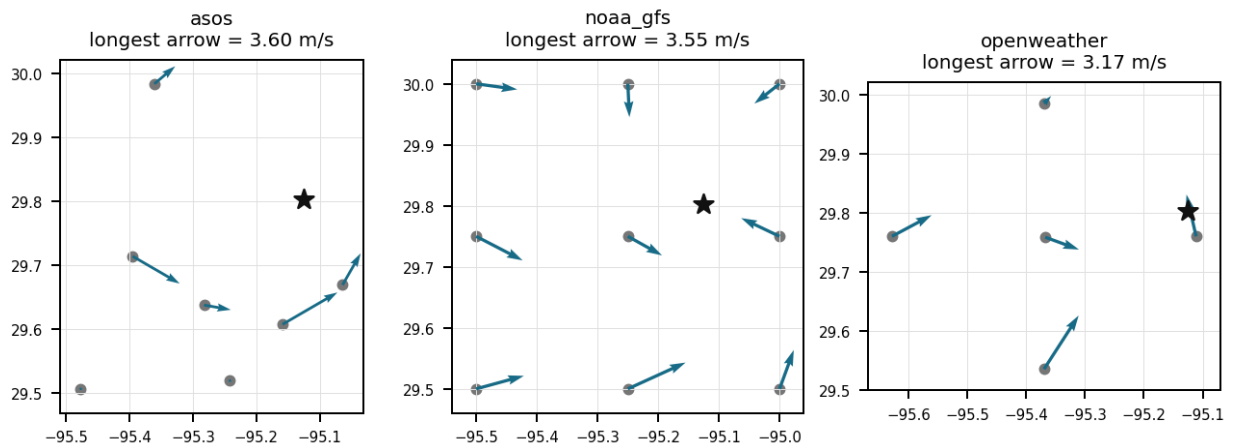
Source	Metric	Values
asos	Relative humidity (humidity)	T41 (15.984 km) 67.85, EFD (21.962 km) 64.01, HOU (23.811 km) 65.15, MCJ (27.821 km) 60.77, IAH (30.369 km) 64.28, LVJ (33.494 km) 69.93, AXH (47.335 km) 68.13
asos	Air temperature (temperature)	T41 (15.984 km) 30.79, EFD (21.962 km) 31.48, HOU (23.811 km) 31.16, MCJ (27.821 km) 31.61, IAH (30.369 km) 31.16, LVJ (33.494 km) 30.23, AXH (47.335 km) 30.21
asos	Wind direction (wind_direction)	T41 (15.984 km) 190.6, EFD (21.962 km) 192.7, HOU (23.811 km) 198.9, MCJ (27.821 km) 208.4, IAH (30.369 km) 212.4, LVJ (33.494 km) 187.1, AXH (47.335 km) 166.4
asos	Wind speed (wind_speed)	T41 (15.984 km) 3.159, EFD (21.962 km) 3.511, HOU (23.811 km) 3.53, MCJ (27.821 km) 3.723, IAH (30.369 km) 2.687, LVJ (33.494 km) 2.004, AXH (47.335 km) 1.536
noaa_gfs	500-hPa geopotential height (gh_500)	gfs:29.75,-95.25 (13.368 km) 5930, gfs:29.75,-95.00 (13.456 km) 5929, gfs:30.00,-95.25 (25.007 km) 5932, gfs:30.00,-95.00 (25.054 km) 5931, gfs:29.50,-95.25 (35.743 km) 5928, gfs:29.50,-95.00 (35.777 km) 5927, gfs:29.75,-95.50 (36.615 km) 5931, gfs:30.00,-95.50 (42.242 km) 5933, +1 more
noaa_gfs	Planetary boundary layer height (pbl_height)	gfs:29.75,-95.25 (13.368 km) 886.1, gfs:29.75,-95.00 (13.456 km) 573.5, gfs:30.00,-95.25 (25.007 km) 856.4, gfs:30.00,-95.00 (25.054 km) 407.9, gfs:29.50,-95.25 (35.743 km) 508.8, gfs:29.50,-95.00 (35.777 km) 522.4, gfs:29.75,-95.50 (36.615 km) 959.3, gfs:30.00,-95.50 (42.242 km) 863, +1 more

Source	Metric	Values
noaa_gfs	Precipitable water (precipitable_water)	gfs:29.75,-95.25 (13.368 km) 43.31, gfs:29.75,-95.00 (13.456 km) 44.39, gfs:30.00,-95.25 (25.007 km) 43.03, gfs:30.00,-95.00 (25.054 km) 43.72, gfs:29.50,-95.25 (35.743 km) 44.15, gfs:29.50,-95.00 (35.777 km) 44.71, gfs:29.75,-95.50 (36.615 km) 42.29, gfs:30.00,-95.50 (42.242 km) 42.36, +1 more
noaa_gfs	Surface pressure (surface_pressure)	gfs:29.75,-95.25 (13.368 km) 1009, gfs:29.75,-95.00 (13.456 km) 1010, gfs:30.00,-95.25 (25.007 km) 1008, gfs:30.00,-95.00 (25.054 km) 1008, gfs:29.50,-95.25 (35.743 km) 1009, gfs:29.50,-95.00 (35.777 km) 1010, gfs:29.75,-95.50 (36.615 km) 1008, gfs:30.00,-95.50 (42.242 km) 1006, +1 more
noaa_gfs	850-hPa temperature (t_850)	gfs:29.75,-95.25 (13.368 km) 22.07, gfs:29.75,-95.00 (13.456 km) 22.01, gfs:30.00,-95.25 (25.007 km) 22.1, gfs:30.00,-95.00 (25.054 km) 22.17, gfs:29.50,-95.25 (35.743 km) 21.96, gfs:29.50,-95.00 (35.777 km) 21.97, gfs:29.75,-95.50 (36.615 km) 22.12, gfs:30.00,-95.50 (42.242 km) 22.25, +1 more
noaa_gfs	10-m eastward wind (u_10m)	gfs:29.75,-95.25 (13.368 km) 0.9858, gfs:29.75,-95.00 (13.456 km) 0.4441, gfs:30.00,-95.25 (25.007 km) 0.685, gfs:30.00,-95.00 (25.054 km) 0.6116, gfs:29.50,-95.25 (35.743 km) 1.122, gfs:29.50,-95.00 (35.777 km) 1.38, gfs:29.75,-95.50 (36.615 km) 1.381, gfs:30.00,-95.50 (42.242 km) 1.624, +1 more
noaa_gfs	10-m northward wind (v_10m)	gfs:29.75,-95.25 (13.368 km) -0.3088, gfs:29.75,-95.00 (13.456 km) -0.1055, gfs:30.00,-95.25 (25.007 km) -0.9463, gfs:30.00,-95.00 (25.054 km) -0.838, gfs:29.50,-95.25 (35.743 km) 0.2687, gfs:29.50,-95.00 (35.777 km) 0.5695, gfs:29.75,-95.50 (36.615 km) -0.4971, gfs:30.00,-95.50 (42.242 km) -0.7271, +1 more
openaq	Ozone (ozone)	3214 (0.023 km) 0.03111, 2800 (6.25 km) 0.02638, 2039 (9.95 km) 0.02793, 339 (13.864 km) 0.03294, 332 (14.755 km) 0.03344, 3378 (14.86 km) 0.02898, 2046 (15.568 km) 0.02729, 325 (20.836 km) 0.009631, +8 more
openaq	PM10 (pm10)	271 (14.755 km) 52.58, 13380082 (14.86 km) 45.94, 15852419 (18.937 km) 35.79
openaq	PM2.5 (pm25)	13787195 (9.338 km) 12.01, 14157976 (13.348 km) 10.65, 9201803 (14.208 km) 10.42, 9489522 (14.512 km) 9.361, 268 (14.755 km) 17.91, 3379 (14.86 km) 13.28, 15539682 (14.878 km) 10.23, 8967123 (14.906 km) 11.31, +68 more
openweather	Cloud cover (cloud_cover)	grid:east (4.904 km) 22.54, grid:center (23.876 km) 24.3, grid:north (31.073 km) 17.32, grid:south (37.919 km) 28.54, grid:west (48.77 km) 21.88
openweather	Relative humidity (humidity)	grid:east (4.904 km) 73.58, grid:center (23.876 km) 65.8, grid:north (31.073 km) 66.6, grid:south (37.919 km) 69.54, grid:west (48.77 km) 67.76
openweather	Precipitation rate (precipitation)	grid:east (4.904 km) 0.2261, grid:center (23.876 km) 0.07761, grid:north (31.073 km) 0.01292, grid:south (37.919 km) 0.2347, grid:west (48.77 km) 0.03764
openweather	Surface pressure (pressure)	grid:east (4.904 km) 1011, grid:center (23.876 km) 1011, grid:north (31.073 km) 1011, grid:south (37.919 km) 1011, grid:west (48.77 km) 1011
openweather	Air temperature (temperature)	grid:east (4.904 km) 31.15, grid:center (23.876 km) 32.42, grid:north (31.073 km) 31.69, grid:south (37.919 km) 31.29, grid:west (48.77 km) 31.72
openweather	Wind direction (wind_direction)	grid:east (4.904 km) 207.7, grid:center (23.876 km) 228.7, grid:north (31.073 km) 183.5, grid:south (37.919 km) 206.2, grid:west (48.77 km) 198.3
openweather	Wind speed (wind_speed)	grid:east (4.904 km) 2.628, grid:center (23.876 km) 2.108, grid:north (31.073 km) 2.012, grid:south (37.919 km) 2.287, grid:west (48.77 km) 1.508
purpleair	PM2.5 (pm25)	186271 (3.011 km) 14.85, 148709 (3.649 km) 13.16, 166441 (3.977 km) 12.68, 243843 (5.149 km) 15.32, 186277 (5.913 km) 14.27, 268171 (9.328 km) 11.46, 213701 (9.828 km) 11.99, 98633 (11.208 km) 3.674, +58 more
tceq	Carbon monoxide (co)	482011039 (14.787 km) 0.1016, 482011035 (14.878 km) 0.1677, 482011052 (25.342 km) 0.1727
tceq	Nitrogen dioxide (no2)	482010026 (0.0 km) 6.934, 482011015 (6.597 km) 9.585, 482011039 (14.787 km) 7.139, 482011035 (14.878 km) 7.009, 482010416 (20.843 km) 6.222, 482010024 (22.224 km) 0.9467, 482011052 (25.342 km) 5.062, 482011050 (26.633 km) 3.402, +4 more
tceq	Sulfur dioxide (so2)	482011039 (14.787 km) 0.5, 482011035 (14.878 km) -0.2933, 482010051 (39.113 km) -0.006452

Ground observation locations in the stored 72-hour context
dashed rings are great-circle distance from the anomaly



Event-nearest wind vectors from stored values (arrow points downwind)



Claims

Mark exactly one box per claim: V (Valid), I (Invalid), or U (Unsure). To skip a claim, leave all three boxes blank; a blank is recorded as missing and never turned into Unsure. Each claim appears exactly once, and the number printed next to it is how I'll refer to that claim if I need to ask you about it.

Label definitions, from the labeling guide

Valid (V): the claim holds up at the precision it states, and the evidence in the packet supports it.

Invalid (I): the evidence in the packet contradicts the claim, the claim misstates a measurement, or the physical reasoning does not hold.

Unsure (U): you cannot tell from what is shown. This covers thin evidence, measurements that cannot resolve the claim, ambiguous wording, and anything outside what you are comfortable judging.

Missing or thin evidence means Unsure, not Invalid.

A cause that is plausible but unproven stays Unsure unless the evidence actually contradicts it.

Some claims bundle several statements into one sentence. Judge those by their parts. If every part earns the same verdict, use that verdict.

Claim 1 of 35

The PM2.5 levels exceeded the normal range, with a peak value of 14.85 ug/m3 at sensor 186271.

Mark exactly one:	☐ V	☐ I	☐ U
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Note (optional):

Claim 2 of 35

The air quality anomaly in Houston, Texas on July 22nd was caused by a combination of factors including high temperatures, low wind speeds, and nearby industrial activities.

Mark exactly one:	☐ V	☐ I	☐ U
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Note (optional):

Claim 3 of 35

The 6-hour mean across tceq NO2 monitors for 2026-07-22 00Z was 10.57 ppb, which is much lower than the 44.3 ppb reading at the anomaly monitor at 2026-07-22T02:00:00+00:00.

Mark exactly one:

☐ V

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Note (optional):

Claim 4 of 35

The air quality anomaly in Houston, Texas is caused by high levels of PM2.5.

Mark exactly one:

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Note (optional):

Claim 5 of 35

Local dispersion near the anomaly was only modest because nearby surface winds were light at about 1.8 to 2.1 m/s, while the modeled planetary boundary layer height near 00Z was about 1536 m rather than strongly collapsed.

Mark exactly one:

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Note (optional):

Claim 6 of 35

Meteorological support for accumulation is moderate rather than strong because surface winds near the anomaly were light at about 1.8 to 2.1 m/s with very low cloud cover and no precipitation, but the gfs planetary boundary layer height nearest in time was still about 1536 m at 00Z, which does not indicate an already well-established very shallow nocturnal boundary layer at the exact event time.

Mark exactly one:

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Note (optional):

Claim 7 of 35

Regional average NO₂ levels across TCEQ monitoring stations peaked during the 6-hour window centered on 2026-07-22T00:00:00+00:00 at 10.57 ppb before returning to baseline levels around 6.37 ppb in the subsequent period.

Mark exactly one:

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Note (optional):

Claim 8 of 35

The anomaly occurred on July 22nd, around 01:59:46 UTC, at a location approximately 14.959 km from the query point.

Mark exactly one:

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Note (optional):

Claim 9 of 35

A surface NO₂ concentration peak of 44.3 ppb was recorded by a TCEQ monitoring station in Houston at 02:00 UTC on July 22, 2026.

Mark exactly one:☐ V☐ I☐ U**Note (optional):****Claim 10 of 35**

A broad regional photochemical or wildfire episode is contradicted because ozone near the anomaly was only 0.042 ppm at 02:00 UTC, particulate matter was only modestly elevated near the anomaly, and the nearest prior-day satellite column fields did not show unusually high NO₂ or CO over the area before the spike.

Mark exactly one:☐ V☐ I☐ U**Note (optional):****Claim 11 of 35**

A broad transported smoke or regional combustion episode is a less plausible primary cause of the NO₂ spike because satellite CO and NO₂ columns before the event were not unusually high, surface PM_{2.5} and PM₁₀ were only moderate around the event time, and no precipitation or cloud-driven convective disturbance was present to suggest a large-scale plume passage.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 12 of 35

Regional wildfire or broad-area secondary pollution is unlikely because concurrent CO and SO₂ near the area remained low, PM_{2.5} was only moderate, ozone was not unusually high for Houston, and Sentinel-5P did not show an elevated regional NO₂ column at the next overpass.

Mark exactly one:☐ V☐ I☐ U**Note (optional):****Claim 13 of 35**

The TCEQ NO₂ monitor at 29.803, -95.126 recorded 44.3 ppb at 2026-07-22T02:00:00+00:00, which is an extreme local spike relative to the expected 6.78 ppb background and the 07-22 00Z 6-hour multi-monitor mean of 10.57 ppb.

Mark exactly one:☐ V☐ I☐ U**Note (optional):****Claim 14 of 35**

The air quality anomaly is related to PM_{2.5} levels.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 15 of 35

Atmospheric trapping was facilitated by light surface winds recorded around 2.05 m/s by ASOS weather stations along with low night-time planetary boundary layer heights modeled by GFS during the anomaly period.

Mark exactly one:☐ V☐ I☐ U**Note (optional):****Claim 16 of 35**

The air quality anomaly in Houston, Texas is not caused by wind direction or speed.

Mark exactly one:☐ V☐ I☐ U**Note (optional):****Claim 17 of 35**

The air quality anomaly in Houston, Texas is not caused by changes in temperature.

Mark exactly one:☐ V☐ I☐ U**Note (optional):****Claim 18 of 35**

A nearby industrial facility or power plant is releasing excessive amounts of particulate matter and volatile organic compounds into the atmosphere.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 19 of 35

Mobile source emissions from evening traffic contributed to the localized spike in NO₂ concentration detected at the TCEQ monitoring site.

Mark exactly one:☐ V☐ I☐ U**Note (optional):****Claim 20 of 35**

Weak near-surface ventilation and nighttime retention likely amplified the NO₂ concentration at the monitor because local winds were light at about 1.8 to 2.1 m/s near the event and nocturnal boundary layers in the period commonly collapsed to roughly 100 to 200 m later in the night, conditions that favor short-lived accumulation of primary NO_x near sources.

Mark exactly one:☐ V☐ I☐ U**Note (optional):****Claim 21 of 35**

A combination of high-pressure weather systems and stagnant air patterns is trapping pollutants in the Houston area, leading to poor air quality.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 22 of 35

TCEQ ground monitoring recorded a severe nitrogen dioxide spike reaching 44.3 ppb at 02:00 UTC on July 22, 2026, contrasting with the regional network baseline mean of approximately 6.23 ppb.

Mark exactly one:

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Note (optional):

Claim 23 of 35

A severe NO₂ concentration spike reaching 44.3 ppb was observed at coordinates (29.803, -95.126) on 2026-07-22T02:00:00+00:00, compared to an expected mean value of 6.78 ppb and a z-score of 8.90.

Mark exactly one:

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Note (optional):

Claim 24 of 35

The temperature inversion over the Houston area is causing a buildup of pollutants near the surface.

Mark exactly one:

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Note (optional):

Claim 25 of 35

Local industrial combustion sources in the Houston region released nitrogen oxides that accumulated near the TCEQ monitoring station under weak surface winds.

Mark exactly one:

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Note (optional):

Claim 26 of 35

Sentinel-5P satellite observations recorded stable total tropospheric NO₂ column densities between July 21 and July 22, 2026, indicating that the NO₂ enhancement was confined to the immediate surface boundary layer rather than a broad regional atmospheric plume.

Mark exactly one:

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Note (optional):

Claim 27 of 35

The tceq NO₂ anomaly at 29.803, -95.126 had an expected value of 6.782638888888889 ppb, a z-score of 8.899715622862008, and a severity classification of severe.

Mark exactly one:

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Note (optional):

Claim 28 of 35

The tceq NO₂ monitor at 29.803, -95.126 recorded 44.3 ppb at 2026-07-22T02:00:00+00:00.

Mark exactly one:

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Note (optional):

Claim 29 of 35

The PM_{2.5} levels are elevated, with a mean of 10.7511 ug/m³ and a maximum value of 149.3 ug/m³.

Mark exactly one:

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Note (optional):

Claim 30 of 35

ASOS weather stations recorded light surface winds near 2.0 m/s during the time of the NO₂ anomaly, limiting horizontal dispersion of local emissions.

Mark exactly one:☐ V☐ I☐ U**Note (optional):****Claim 31 of 35**

A decaying planetary boundary layer height combined with low wind speeds during the evening transition trapped surface emissions and restricted atmospheric dispersion.

Mark exactly one:☐ V☐ I☐ U**Note (optional):****Claim 32 of 35**

A fresh, highly localized NO_x emission plume from nearby combustion sources such as ship-channel industry, port activity, diesel traffic, rail, or nonroad equipment is a leading physical cause of the 44.3 ppb NO₂ spike because the anomaly was extreme at one tceq monitor while regional co-pollutants were only modest and area-wide ozone was not elevated enough to indicate a broad regional event.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 33 of 35

A localized fresh-combustion plume is strongly supported because the tceq NO2 concentration at monitor 482010026 reached 44.3 ppb at 2026-07-22T02:00:00+00:00 while the 6-hour multi-monitor mean was only 10.57 ppb and nearby co was only 0.2 ppm with so2 at 0.4 ppb, a pattern consistent with a short-lived local traffic or combustion source rather than a region-wide sulfur-rich plume.

Mark exactly one:☐ V☐ I☐ U**Note (optional):****Claim 34 of 35**

The GFS model indicated a collapse of the planetary boundary layer height during the late evening hours on July 21 to July 22, 2026, restricting vertical atmospheric mixing.

Mark exactly one:☐ V☐ I☐ U**Note (optional):****Claim 35 of 35**

NO2 levels were also elevated, with a peak value of 44.3 ppb at monitor 482010026.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Optional: most likely cause

Your best guess at the true cause of this event, in your own words. "Insufficient evidence to determine a single cause" is a perfectly fine answer.